# Switch Core 1.0.0

## JUnit Test Execution Report

RuPay Credit and Prepaid Cards including NCMC

| | |
|---|---|
| Report reference | SC-1.0.0-JUNIT-20261016 |
| Execution completed | 16 October 2026, 06:28:00 IST |
| Command / result | mvn clean test / BUILD SUCCESS (exit code 0) |
| Source baseline | a33b3a4e97a468c99228d2386381201ecc7c652d plus local source and test changes |
| Toolchain | Maven 3.9.10; Homebrew OpenJDK 25.0.2; macOS aarch64 |
| Execution duration | 25.378 seconds |

### Executive summary

The Maven test lifecycle completed successfully. Of 961 reported tests, 945 passed and 16 were skipped. There were no assertion failures or execution errors. The skipped database integration tests were not executed and are not counted as passed.

| Reported | Passed | Failures | Errors | Skipped |
|---|---|---|---|---|
| 961 | 945 | 0 | 0 | 16 |

This report is vendor verification evidence for the source code handover. The automated suite spans the shared project repository; these totals are not an isolated RuPay product certification result. Client UAT, scheme certification, live dependency integration, and production authorization are separate activities.

# Coverage and execution scope

| Metric | Covered | Total | Coverage |
|---|---|---|---|
| Lines | 9,598 | 11,721 | 81.89% |
| Branches | 4,311 | 6,619 | 65.13% |

Coverage is taken from the JaCoCo aggregate report and reflects its configured exclusions. The test lifecycle generates this report but does not run the verify-phase coverage gate or package an artifact.

## Skipped database tests

| Suite | Skipped | Reason |
|---|---|---|
| CmsRegistrationPersistenceTest | 7 | CMS_SWITCH_TEST_DB not configured |
| CmsDatabaseIntegrationTest | 9 | CMS_SWITCH_TEST_DB not configured |

The database-backed persistence and integration scenarios above remain unverified in this execution. Local mock-server tests are included; they do not prove connectivity or interoperability with client HSM, CMS, or Accounting environments.

## Source traceability

The tested source fingerprint matches the current local source and test files. The release tag must identify a final commit containing those changes; the base commit alone does not reproduce this run.
SHA-256: a8acaab141a7e5f6c6c4c90b828fe8c68056cd4d16de46bcc38403046cc2df28

Fingerprint method: sort tracked and non-ignored untracked paths under src/ and pom.xml; concatenate each UTF-8 relative path, a null byte, and its binary SHA-256 content digest, then hash the concatenation. This run covers 441 paths. Environment files, archives, generated output, and documentation are excluded.

## Evidence included

suite-results.csv lists all 132 suite results. junit-results.xml contains the machine-readable JUnit result hierarchy with test names, durations, and skip status. Embedded logs and environment properties have been omitted. The release notes and vendor test record refer to report SC-1.0.0-JUNIT-20261016.

# Results by test suite

Columns show reported tests, passed tests, failures, errors, and skipped tests. Fully qualified suite names and timings are available in the accompanying CSV and XML.

| Test suite | Tests | Pass | Fail | Error | Skip |
|---|---|---|---|---|---|
| CircularDependencyFixTest | 2 | 2 | 0 | 0 | 0 |
| CircularDependencyTest | 2 | 2 | 0 | 0 | 0 |
| cmsintegration.CmsAuthorizationLookupTest | 5 | 5 | 0 | 0 | 0 |
| cmsintegration.CmsAuthorizationPolicyTest | 6 | 6 | 0 | 0 | 0 |
| cmsintegration.CmsCardControllerTest | 3 | 3 | 0 | 0 | 0 |
| cmsintegration.CmsCardIntegrationServiceMutationTest | 4 | 4 | 0 | 0 | 0 |
| cmsintegration.CmsCardIntegrationServiceReadTest | 4 | 4 | 0 | 0 | 0 |
| cmsintegration.CmsCardIntegrationServiceRegistrationTest | 2 | 2 | 0 | 0 | 0 |
| cmsintegration.CmsCardSecurityTest | 7 | 7 | 0 | 0 | 0 |
| cmsintegration.CmsDatabaseIntegrationTest | 9 | 0 | 0 | 0 | 9 |
| cmsintegration.CmsFuturexPinExecutorTest | 2 | 2 | 0 | 0 | 0 |
| cmsintegration.CmsMaterialDecoderTest | 6 | 6 | 0 | 0 | 0 |
| cmsintegration.CmsPinControllerAdditionalTest | 1 | 1 | 0 | 0 | 0 |
| cmsintegration.CmsPinControllerTest | 1 | 1 | 0 | 0 | 0 |
| cmsintegration.CmsPinGrantVerifierTest | 3 | 3 | 0 | 0 | 0 |
| cmsintegration.CmsPinHsmTest | 3 | 3 | 0 | 0 | 0 |
| cmsintegration.CmsPinMaterialDecoderTest | 8 | 8 | 0 | 0 | 0 |
| cmsintegration.CmsPinOperationServiceCoverageTest | 9 | 9 | 0 | 0 | 0 |
| cmsintegration.CmsPinOperationServiceTest | 3 | 3 | 0 | 0 | 0 |
| cmsintegration.CmsPinStateAndKeysTest | 8 | 8 | 0 | 0 | 0 |
| cmsintegration.CmsPinTransactionServiceTest | 2 | 2 | 0 | 0 | 0 |
| cmsintegration.CmsProfileRulesTest | 5 | 5 | 0 | 0 | 0 |
| cmsintegration.CmsRegistrationPersistenceTest | 7 | 0 | 0 | 0 | 7 |
| config.AccountingJsonMigrationTest | 2 | 2 | 0 | 0 | 0 |
| config.IssuerActionGatewayContextTest | 1 | 1 | 0 | 0 | 0 |
| config.RawDataLoggingTest | 2 | 2 | 0 | 0 | 0 |
| config.SwitchApiSecurityConfigTest | 6 | 6 | 0 | 0 | 0 |
| controller.CacheManagementControllerTest | 4 | 4 | 0 | 0 | 0 |

| Test suite | Tests | Pass | Fail | Error | Skip |
|---|---|---|---|---|---|
| controller.CardOnboardingControllerTest | 6 | 6 | 0 | 0 | 0 |
| controller.HsmCmdControllerSecurityTest | 1 | 1 | 0 | 0 | 0 |
| controller.HsmCmdControllerTest | 3 | 3 | 0 | 0 | 0 |
| controller.Iso8583ControllerTest | 6 | 6 | 0 | 0 | 0 |
| controller.IssuerMandateRevocationControllerTest | 1 | 1 | 0 | 0 | 0 |
| emv.CardVerificationResultTest | 31 | 31 | 0 | 0 | 0 |
| emv.CardholderVerificationMethodResultsTest | 20 | 20 | 0 | 0 | 0 |
| emv.ChipDataParserTest | 13 | 13 | 0 | 0 | 0 |
| emv.IssuerApplicationDataTest | 12 | 12 | 0 | 0 | 0 |
| emv.TerminalVerificationResultsTest | 3 | 3 | 0 | 0 | 0 |
| emv.TvrDataTest | 2 | 2 | 0 | 0 | 0 |
| exception.ExceptionHandlingTest | 5 | 5 | 0 | 0 | 0 |
| hsm.HsmUtilitiesTest | 5 | 5 | 0 | 0 | 0 |
| hsm.SimpleNettyHsmClientCoverageTest | 21 | 21 | 0 | 0 | 0 |
| mcc.MccConfigServiceTest | 3 | 3 | 0 | 0 | 0 |
| mcc.MccRuleEngineTest | 23 | 23 | 0 | 0 | 0 |
| mcc.MccSupportTypesTest | 11 | 11 | 0 | 0 | 0 |
| model.MessageTypeTest | 5 | 5 | 0 | 0 | 0 |
| notification.client.NotificationClientIntegrationTest | 3 | 3 | 0 | 0 | 0 |
| notification.exception.NotificationExceptionTest | 2 | 2 | 0 | 0 | 0 |
| notification.mapper.DeclinedTransactionNotificationMapperTest | 4 | 4 | 0 | 0 | 0 |
| notification.service.NotificationServiceImplTest | 4 | 4 | 0 | 0 | 0 |
| service.AccountingApiServiceHttpTest | 6 | 6 | 0 | 0 | 0 |
| service.AccountingApiServiceTest | 4 | 4 | 0 | 0 | 0 |
| service.AesGcmCryptoServiceTest | 4 | 4 | 0 | 0 | 0 |
| service.AuthorizationValidationServicesTest | 8 | 8 | 0 | 0 | 0 |
| service.AutomaticReversalServiceTest | 3 | 3 | 0 | 0 | 0 |
| service.CacheInitializationServiceTest | 4 | 4 | 0 | 0 | 0 |
| service.CacheUpdateServiceTest | 3 | 3 | 0 | 0 | 0 |
| service.CanonicalResponseCodeSeedServiceTest | 2 | 2 | 0 | 0 | 0 |

| Test suite | Tests | Pass | Fail | Error | Skip |
|---|---|---|---|---|---|
| service.CardAuthenticationServiceTest | 2 | 2 | 0 | 0 | 0 |
| service.CardDataEncryptionServiceTest | 1 | 1 | 0 | 0 | 0 |
| service.CardDataServiceTest | 14 | 14 | 0 | 0 | 0 |
| service.ChdKeyCacheServiceTest | 3 | 3 | 0 | 0 | 0 |
| service.ChdKeyProvisioningServiceTest | 4 | 4 | 0 | 0 | 0 |
| service.CvmValidationServiceCertificationTest | 5 | 5 | 0 | 0 | 0 |
| service.CvmValidationServiceOfflinePinTest | 5 | 5 | 0 | 0 | 0 |
| service.CvmValidationServiceTest | 19 | 19 | 0 | 0 | 0 |
| service.CvrValidationServiceTest | 9 | 9 | 0 | 0 | 0 |
| service.DuplicateCheckServiceTest | 5 | 5 | 0 | 0 | 0 |
| service.ExternalEndpointMetricsTest | 5 | 5 | 0 | 0 | 0 |
| service.FinancialRequestProcessorTest | 16 | 16 | 0 | 0 | 0 |
| service.HsmKeepAliveServiceTest | 9 | 9 | 0 | 0 | 0 |
| service.Iso8583ProcessorServiceTest | 7 | 7 | 0 | 0 | 0 |
| service.IssuerMandateRevocationLifecycleServiceTest | 3 | 3 | 0 | 0 | 0 |
| service.IssuerMandateRevocationMessageFactoryTest | 1 | 1 | 0 | 0 | 0 |
| service.IssuerMandateRevocationServiceTest | 3 | 3 | 0 | 0 | 0 |
| service.KeyManagementServiceTest | 17 | 17 | 0 | 0 | 0 |
| service.MessageProcessorServiceTest | 18 | 18 | 0 | 0 | 0 |
| service.NettyHsmServiceLoggingTest | 3 | 3 | 0 | 0 | 0 |
| service.NettyHsmServiceTest | 12 | 12 | 0 | 0 | 0 |
| service.NetworkSessionContextHolderTest | 3 | 3 | 0 | 0 | 0 |
| service.NetworkSessionKeyServiceTest | 7 | 7 | 0 | 0 | 0 |
| service.PinChangeServiceTest | 12 | 12 | 0 | 0 | 0 |
| service.PrepaidCardBalanceSnapshotServiceTest | 1 | 1 | 0 | 0 | 0 |
| service.PrepaidLimitValidationServiceTest | 5 | 5 | 0 | 0 | 0 |
| service.ResponseAssemblerTest | 4 | 4 | 0 | 0 | 0 |
| service.ReversalInfrastructureTest | 3 | 3 | 0 | 0 | 0 |
| service.ReversalProcessorServiceTest | 26 | 26 | 0 | 0 | 0 |
| service.ReversalStorageServiceTest | 6 | 6 | 0 | 0 | 0 |

| Test suite | Tests | Pass | Fail | Error | Skip |
|---|---|---|---|---|---|
| service.RoutingHealthCheckServiceTest | 4 | 4 | 0 | 0 | 0 |
| service.ServiceCodeValidationServiceTest | 1 | 1 | 0 | 0 | 0 |
| service.ServiceSupportClassesTest | 3 | 3 | 0 | 0 | 0 |
| service.StandingInstructionAuthorizationValidationServiceTest | 10 | 10 | 0 | 0 | 0 |
| service.StandingInstructionMandateServiceTest | 13 | 13 | 0 | 0 | 0 |
| service.ThreeDsAuthorizationValidationServiceTest | 2 | 2 | 0 | 0 | 0 |
| service.TransactionChdEncryptionServiceTest | 6 | 6 | 0 | 0 | 0 |
| service.TransactionContextTest | 3 | 3 | 0 | 0 | 0 |
| service.TransactionIdGeneratorServiceTest | 3 | 3 | 0 | 0 | 0 |
| service.TransactionOutcomeMetricsTest | 2 | 2 | 0 | 0 | 0 |
| service.TvrValidationServiceTest | 21 | 21 | 0 | 0 | 0 |
| util.CardVerificationMethodTest | 14 | 14 | 0 | 0 | 0 |
| util.EmvDataUpdaterTest | 4 | 4 | 0 | 0 | 0 |
| util.EntryModeTest | 13 | 13 | 0 | 0 | 0 |
| util.ISOUtilTest | 12 | 12 | 0 | 0 | 0 |
| util.Iso8583MessageHandlerTest | 5 | 5 | 0 | 0 | 0 |
| util.MacGenerationTest | 3 | 3 | 0 | 0 | 0 |
| util.RandomCodeGeneratorsTest | 4 | 4 | 0 | 0 | 0 |
| util.ServiceCodeParserTest | 41 | 41 | 0 | 0 | 0 |
| util.rupay.De54dataTest | 3 | 3 | 0 | 0 | 0 |
| validation.SchemeValidatorServiceTest | 5 | 5 | 0 | 0 | 0 |
| validation.StandingInstructionIdentificationTest | 8 | 8 | 0 | 0 | 0 |
| validation.TlvConditionTest | 8 | 8 | 0 | 0 | 0 |
| validation.TlvUtilsTest | 10 | 10 | 0 | 0 | 0 |
| validation.TransactionIdentificationRuleTest | 5 | 5 | 0 | 0 | 0 |
| validation.card.impl.CardValidatorImplTest | 22 | 22 | 0 | 0 | 0 |
| validation.context.De48ContextTest | 9 | 9 | 0 | 0 | 0 |
| validation.factory.TransactionValidatorFactoryTest | 2 | 2 | 0 | 0 | 0 |
| validation.orchestrator.CardDataContextTest | 4 | 4 | 0 | 0 | 0 |
| validation.orchestrator.TransactionValidationOrchestratorAccountIdTest | 8 | 8 | 0 | 0 | 0 |

| Test suite | Tests | Pass | Fail | Error | Skip |
|---|---|---|---|---|---|
| validation.orchestrator.TransactionValidationOrchestratorCashAtPosTest | 5 | 5 | 0 | 0 | 0 |
| validation.orchestrator.TransactionValidationOrchestratorDe48Test | 5 | 5 | 0 | 0 | 0 |
| validation.orchestrator.TransactionValidationOrchestratorFlowTest | 65 | 65 | 0 | 0 | 0 |
| validation.orchestrator.TransactionValidationOrchestratorMiniStatementTest | 1 | 1 | 0 | 0 | 0 |
| validation.orchestrator.TransactionValidationOrchestratorNepalBhutanTest | 6 | 6 | 0 | 0 | 0 |
| validation.orchestrator.TransactionValidationOrchestratorQuickTest | 4 | 4 | 0 | 0 | 0 |
| validation.transaction.impl.AbstractTransactionTypeValidatorTest | 3 | 3 | 0 | 0 | 0 |
| validation.transaction.impl.AtmFinancialTransactionValidatorTest | 19 | 19 | 0 | 0 | 0 |
| validation.transaction.impl.AtmServiceTransactionValidatorTest | 4 | 4 | 0 | 0 | 0 |
| validation.transaction.impl.EcommerceTransactionValidatorTest | 11 | 11 | 0 | 0 | 0 |
| validation.transaction.impl.NcmcTxnValidationAdditionalTest | 6 | 6 | 0 | 0 | 0 |
| validation.transaction.impl.NcmcTxnValidationAfaTest | 2 | 2 | 0 | 0 | 0 |
| validation.transaction.impl.PurchaseTransactionValidatorAdditionalTest | 15 | 15 | 0 | 0 | 0 |
| validation.transaction.impl.PurchaseTransactionValidatorTest | 4 | 4 | 0 | 0 | 0 |